PRACTICE SHEET

1. If $u = \sin^{-1}(x - y)$, $x = 3t$, $y = t^3$, then what is the derivative of u with respect to ?

(a) $3(1-t^2)$ (b) $3(1-t^2)^{-\frac{1}{2}}$

(c) $5(1-t^2)^{-\frac{1}{2}}$ (d) $5(1-t^2)$

2. If $y = x + e^x$, then what is $\frac{d^2x}{dy^2}$ equal to?

(a) e^x (b) $-\frac{e^x}{(1+e^x)^3}$

(c) $-\frac{e^x}{(1+e^x)}$ (d) $-\frac{e^x}{(1+e^x)^2}$

3. If, $x + y = t - \frac{1}{t}, x^2 + y^2 = t^2 + \frac{1}{t^2}$ What is $\frac{dy}{dx}$ equal to?

(a) $\frac{1}{x}$ (b) $-\frac{1}{x}$

(c) $\frac{1}{x^2}$ (d) $-\frac{1}{x^2}$

4. What is the derivative of $\tan^{-1}\left(\frac{2\cos x + 3\sin x}{3\cos x - 2\sin x}\right)$?

(a) 1 (b) $\frac{1}{2}$

(c) $\frac{1}{3}$ (d) 0

5. If $y = \sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right)$, then what is $\frac{dy}{dx}$ equal to?

(a) $\frac{1}{1+x^2}$ (b) $\frac{1}{\sqrt{1+x^2}}$

(c) $\frac{1}{\sqrt{1-x^2}}$ (d) $\frac{1}{1-x^2}$

6. If $y = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$, then what is $\frac{dy}{dx}$ equal to?

(a) $\frac{2}{1+x^2}$ (b) $\frac{2}{\sqrt{1+x^2}}$

(c) $\frac{2}{\sqrt{1-x^2}}$ (d) $\frac{2}{1-x^2}$

7. If $y = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$, then what is $\frac{dy}{dx}$ equal to?

(a) $\frac{2}{1+x^2}$ (b) $\frac{2}{\sqrt{1+x^2}}$

(c) $\frac{2}{\sqrt{1-x^2}}$ (d) $\frac{2}{1-x^2}$

8. If $y = \sin^{-1} x + \sin^{-1}, \sqrt{1-x^2}$ what is $\frac{dy}{dx}$ equal to?

(a) $\cos^{-1} x + \cos^{-1}\sqrt{1-x^2}$ (b) $\frac{1}{\cos x} + \frac{1}{\cos\sqrt{1-x^2}}$

(c) $\frac{\pi}{2}$ (d) 0

9. If $f(x) = \log_e [\log_e x]$, then what is f' (e) equal to?

(a) e^{-1} (b) e

(c) 1 (d) 0

10. For the curve $\sqrt{x} + \sqrt{y} = 1$, what is the value of $\frac{dy}{dx}$ at $\left(\frac{1}{4}, \frac{1}{4}\right)$?

(a) $\frac{1}{2}$ (b) 1

(c) –1 (d) 2

11. If $y = \frac{1}{\log_{10} x}$, then what is $\frac{dy}{dx}$ equal to

(a) x (b) $x \log_e 10$

(c) $(\log_e 10)^2 (\log 10^e)$ (d) $-\frac{\log_e 10}{x}$